

Conjugate Gradient Method

Recurrence Relation: $\mathbf{x}^{(k)} = \mathbf{x}^{(k-1)} + \alpha^{(k)} \cdot \mathbf{d}^{(k)}$, $\mathbf{d}$ is search direction.

Residual Vector: $\mathbf{r}^{(k)} = \mathbf{b} - \mathbf{A}\mathbf{x}^{(k)}$; $k = 0$
 $\mathbf{r}^{(k)} = \mathbf{r}^{(k-1)} - \alpha^{(k)} \cdot \mathbf{A}\mathbf{d}^{(k)}$; $k = 1, 2, 3, \dots$ $\alpha^{(k)} = \frac{\langle \mathbf{r}^{(k-1)}, \mathbf{r}^{(k-1)} \rangle}{\langle \mathbf{d}^{(k)}, \mathbf{A}\mathbf{d}^{(k)} \rangle}$; $k = 1, 2, 3, \dots$

Search Directions of Conjugate Gradient Method:

$$\mathbf{d}^{(1)} = \mathbf{r}^{(0)}$$

$\{\mathbf{d}^{(1)}, \mathbf{d}^{(2)}, \dots, \mathbf{d}^{(n)}\}$ are $\mathbf{A}$ -orthogonal set. $\rightarrow \langle \mathbf{d}^{(i)}, \mathbf{A}\mathbf{d}^{(j)} \rangle = 0, i \neq j$

Construction of Search Directions:

$$\mathbf{d}^{(k+1)} = \mathbf{r}^{(k)} + \beta^{(k)} \cdot \mathbf{d}^{(k)}$$

$$\beta^{(k)} = \frac{\langle \mathbf{r}^{(k)}, \mathbf{r}^{(k)} \rangle}{\langle \mathbf{r}^{(k-1)}, \mathbf{r}^{(k-1)} \rangle}$$

Conjugate Gradient Method

$$\mathbf{A} = \begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix}, \mathbf{b} = \begin{Bmatrix} 2 \\ -8 \end{Bmatrix}, \mathbf{x}^{(0)} = \begin{Bmatrix} -2 \\ -2 \end{Bmatrix}$$

$k = 1$

$$\mathbf{r}^{(k-1)} = \mathbf{r}^{(0)} = \mathbf{b} - \mathbf{A}\mathbf{x}^{(0)} = (12, 8), \mathbf{d}^{(1)} = \mathbf{r}^{(0)}$$

$$\alpha^{(k)} = \alpha^{(1)} = \frac{\langle \mathbf{r}^{(0)}, \mathbf{r}^{(0)} \rangle}{\langle \mathbf{d}^{(1)}, \mathbf{A}\mathbf{d}^{(1)} \rangle} = \frac{208}{1200} = 0.173333$$

$$\left. \begin{array}{l} \mathbf{x}^{(k)} = \mathbf{x}^{(1)} = \mathbf{x}^{(0)} + \alpha^{(1)} \cdot \mathbf{d}^{(1)} \\ = (-2, -2) + 0.173333 \cdot (12, 8) \\ = (0.08, -0.613333) \end{array} \right\}$$

$$\begin{aligned} \mathbf{r}^{(1)} &= \mathbf{r}^{(0)} - \alpha^{(1)} \cdot \mathbf{A}\mathbf{d}^{(1)} \\ &= (12, 8) - 0.173333 \cdot (52, 72) \\ &= (2.986666, -4.48) \end{aligned}$$

$$\begin{aligned} \varepsilon^{(1)} &= \frac{\|\mathbf{r}^{(1)}\|_2}{\|\mathbf{r}^{(0)}\|_2} = 0.373333 \\ \varepsilon^{(1)} &> 10^{-6} \rightarrow \text{Continue} \end{aligned}$$

$$\beta^{(1)} = \frac{\langle \mathbf{r}^{(1)}, \mathbf{r}^{(1)} \rangle}{\langle \mathbf{r}^{(0)}, \mathbf{r}^{(0)} \rangle} = \frac{5.384290}{14.422205} = 0.139378$$

$k = 2$

$$\begin{aligned} \mathbf{d}^{(k)} = \mathbf{d}^{(2)} &= \mathbf{r}^{(1)} + \beta^{(1)} \cdot \mathbf{d}^{(1)} \\ &= (2.986666, -4.48) + 0.139378 \cdot (12, 8) \\ &= (4.6592, -3.364978) \end{aligned}$$

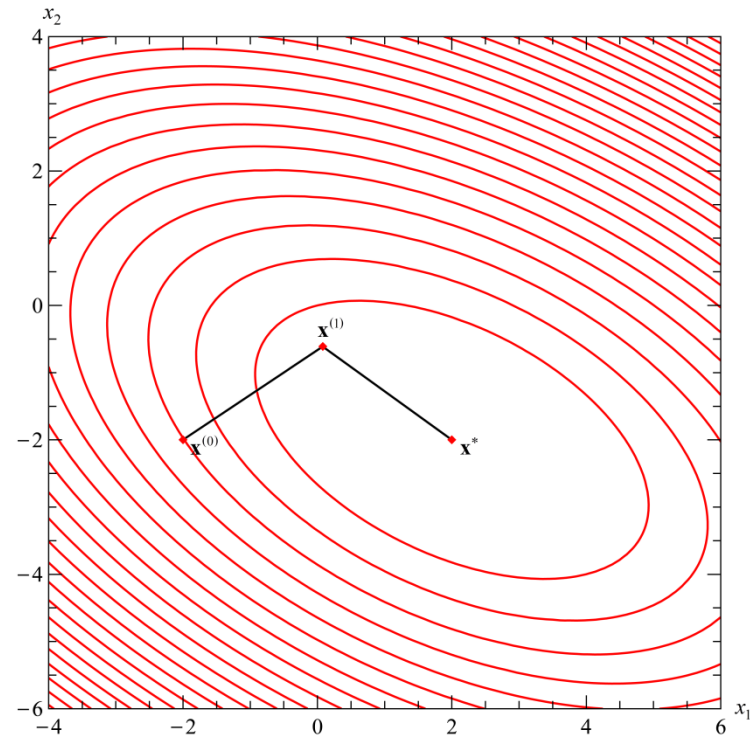
$$\alpha^{(k)} = \alpha^{(2)} = \frac{\langle \mathbf{r}^{(1)}, \mathbf{r}^{(1)} \rangle}{\langle \mathbf{d}^{(2)}, \mathbf{A}\mathbf{d}^{(2)} \rangle} = \frac{5.384290}{70.350469} = 0.412088$$

$$\left. \begin{array}{l} \mathbf{x}^{(k)} = \mathbf{x}^{(2)} = \mathbf{x}^{(1)} + \alpha^{(2)} \cdot \mathbf{d}^{(2)} \\ = (0.08, -0.613333) + 0.412088 \cdot (4.6592, -3.364978) \\ = (2, -2) \end{array} \right\}$$

$$\begin{aligned} \mathbf{r}^{(2)} &= \mathbf{r}^{(1)} - \alpha^{(2)} \cdot \mathbf{A}\mathbf{d}^{(2)} \\ &= (2.986666, -4.48) - 0.412088 \cdot (7.247644, -10.871467) \\ &= (0, 0) \end{aligned}$$

$$\begin{aligned} \varepsilon^{(2)} &= \frac{\|\mathbf{r}^{(2)}\|_2}{\|\mathbf{r}^{(0)}\|_2} = 0 \\ \varepsilon^{(2)} &< 10^{-6} \rightarrow \text{Stop} \end{aligned}$$

Conjugate Gradient Method



Conjugate Gradient Method for Normal Equations

$$\mathbf{Ax} = \mathbf{b} \rightarrow \mathbf{A}^T \mathbf{Ax} = \mathbf{A}^T \mathbf{b} \quad \mathbf{A}^T \mathbf{A} \text{ is symmetric and positive-semidefinite for any } \mathbf{A}.$$
$$\langle \mathbf{x}, \mathbf{Ax} \rangle \geq 0$$

Recurrence Relation: $\mathbf{x}^{(k)} = \mathbf{x}^{(k-1)} + \alpha^{(k)} \cdot \mathbf{d}^{(k)}$, $\mathbf{d}$ is search direction.

Residual Vector: $\mathbf{r}^{(k)} = \mathbf{b} - \mathbf{Ax}^{(k)}$; $k = 0$

$$\mathbf{r}^{(k)} = \mathbf{r}^{(k-1)} - \alpha^{(k)} \cdot \mathbf{Ad}^{(k)}; k = 1, 2, 3, \dots$$
$$\alpha^{(k)} = \frac{\langle \mathbf{A}^T \mathbf{r}^{(k-1)}, \mathbf{A}^T \mathbf{r}^{(k-1)} \rangle}{\langle \mathbf{Ad}^{(k)}, \mathbf{Ad}^{(k)} \rangle}; k = 1, 2, 3, \dots$$

Search Directions of Conjugate Gradient Method for Normal Equations:

$$\mathbf{d}^{(1)} = \mathbf{A}^T \mathbf{r}^{(0)}$$

$$\{\mathbf{d}^{(1)}, \mathbf{d}^{(2)}, \dots, \mathbf{d}^{(n)}\} \text{ are } \mathbf{A}\text{-orthogonal set.} \rightarrow \langle \mathbf{d}^{(i)}, \mathbf{Ad}^{(j)} \rangle = 0, i \neq j$$

Construction of Search Directions:

$$\mathbf{d}^{(k+1)} = \mathbf{A}^T \mathbf{r}^{(k)} + \beta^{(k)} \cdot \mathbf{d}^{(k)}$$

$$\beta^{(k)} = \frac{\langle \mathbf{A}^T \mathbf{r}^{(k)}, \mathbf{A}^T \mathbf{r}^{(k)} \rangle}{\langle \mathbf{A}^T \mathbf{r}^{(k-1)}, \mathbf{A}^T \mathbf{r}^{(k-1)} \rangle}$$

Preconditioned Conjugate Gradient Method for Normal Equations

Recurrence Relation: $\mathbf{x}^{(k)} = \mathbf{x}^{(k-1)} + \alpha^{(k)} \cdot \mathbf{d}^{(k)}$, $\mathbf{d}$ is search direction.

Residual Vector: $\mathbf{r}^{(k)} = \mathbf{b} - \mathbf{A}\mathbf{x}^{(k)}$; $k = 0$

$$\mathbf{r}^{(k)} = \mathbf{r}^{(k-1)} - \alpha^{(k)} \cdot \mathbf{A}\mathbf{d}^{(k)}; k = 1, 2, 3 \quad \alpha^{(k)} = \frac{\langle \mathbf{z}^{(k-1)}, \mathbf{A}^T \mathbf{r}^{(k-1)} \rangle}{\langle \mathbf{A}\mathbf{d}^{(k)}, \mathbf{A}\mathbf{d}^{(k)} \rangle}; k = 1, 2, 3, \dots$$

Preconditioned Vector: $\mathbf{z}^{(k)} = \mathbf{M}^{-1} \mathbf{A}^T \mathbf{r}^{(k)}$; $k = 0, 1, 2, 3, \dots$ $\mathbf{M}$ is diagonal matrix.

Search Directions of PCGNE:

$$\mathbf{d}^{(1)} = \mathbf{z}^{(0)}$$

$\{\mathbf{d}^{(1)}, \mathbf{d}^{(2)}, \dots, \mathbf{d}^{(n)}\}$ are $\mathbf{A}$ -orthogonal set. $\rightarrow \langle \mathbf{d}^{(i)}, \mathbf{A}\mathbf{d}^{(j)} \rangle = 0, i \neq j$

Construction of Search Directions:

$$\mathbf{d}^{(k+1)} = \mathbf{z}^{(k)} + \beta^{(k)} \cdot \mathbf{d}^{(k)}$$

$$\beta^{(k)} = \frac{\langle \mathbf{z}^{(k)}, \mathbf{A}^T \mathbf{r}^{(k)} \rangle}{\langle \mathbf{z}^{(k-1)}, \mathbf{A}^T \mathbf{r}^{(k-1)} \rangle}$$